Fairness in Job Recommender Systems: Balancing Accuracy and Equal Opportunity

Efe Karatas (15773744), Zi Cheng Sunny Li (15708802), Sharjeel Mohammad (15270785) & Samim Amdam (15266931)

Abstract

This study investigates whether job recommender systems distribute economic opportunities fairly across user groups. Using a dataset of 1,417,471 job applications, we trained three models: an Alternating Least Squares (ALS) baseline, a hybrid model, and a fairness-aware re-ranking model. Fairness was operationalised as the exposure gap to high-opportunity jobs between users with and without managerial experience. Results show that standard models reproduce existing inequalities, whereas a small fairness parameter ($\beta = 0.002$) reduces the exposure gap from 0.006 to 0.001 without significantly harming recommendation quality (nDCG10). These findings confirm that targeted, carefully tuned fairness interventions can improve opportunity distribution in job recommender systems with minimal accuracy cost.

I. Introduction

Online labor market platforms such as LinkedIn and Indeed rely on recommender systems to filter and rank job vacancies, shaping which opportunities users see (Çelik Ertuğrul & Bitirim, 2025; Dillahunt et al., 2021; Zheng et al., 2012). By determining visibility, these systems influence how economic opportunities are distributed, as users typically engage with only a small portion of ranked results (de Ruijt & Bhulai, 2021).

Because they are trained on historical interaction data, recommender systems may reproduce or amplify existing labor market inequalities (Kumar et al., 2023; Raghavan et al., 2020). This raises concerns about fairness and accountability (Jin et al., 2023; Wang et al., 2022). However, it remains unclear to what extent such systems reinforce inequalities and whether fairness-aware methods can mitigate these effects without sacrificing accuracy.

This paper investigates the following research question:

> *Do job recommender systems distribute economic opportunities fairly across different groups, or do they reproduce existing structural inequality?*

To study this question, we construct a simulated labor market in which structural bias is embedded in the interaction data. Our goal is to analyze whether standard recommender models reproduce existing exposure inequalities and whether fairness aware methods can reduce these disparities, potentially at the cost of some accuracy.

II. Theoretical Background

A. Recommender Systems in the Labor Market

Online labor market platforms increasingly use recommender systems to match job seekers with relevant vacancies and employers with suitable candidates, helping reduce information overload by filtering and ranking postings (Dillahunt et al., 2021; Reusens et al., 2018).

Job recommender systems operate in a two sided environment, requiring mutual interest between job seekers and employers, unlike traditional systems that recommend passive items (de Ruijt & Bhulai, 2021).

They employ approaches such as collaborative filtering, content based methods, and hybrid models that integrate user profiles, job data, and interaction histories, thereby shaping how opportunities are discovered on digital labor platforms (de Ruijt & Bhulai, 2021).

B. Ranking, Visibility, and Exposure

In recommender systems, items are shown in ranked lists where higher positions receive significantly more attention, making ranking a key determinant of visibility and user interaction (Singh & Joachims, 2018).

In job recommender systems, exposure is especially important because it determines which opportunities are seen by job seekers (Reusens et al., 2018). Since users engage with only a small subset of results, ranking algorithms shape how opportunities are distributed across jobs and users, potentially leading to unequal attention for certain jobs or groups (Geyik et al., 2019; Kumar et al., 2023).

Therefore, recent research highlights the need to account for exposure when evaluating recommender systems, particularly in domains like employment where recommendations affect access to important opportunities (Wang et al., 2022).

C. Algorithmic Bias and Structural Inequality

Algorithmic decision making systems in hiring and job recommendation are often trained on historical data that reflect existing inequalities in the labor market. As a result, recommender systems may reproduce or amplify patterns of discrimination that already exist in society (Kumar et al., 2023; Raghavan et al., 2020). Structural inequalities related to gender, race, disability, and socioeconomic background are frequently embedded in recruitment data and

evaluation processes, which can lead to systematic disadvantages for certain groups when these data are used to train algorithms.

In the hiring domain, biases can arise from multiple sources, including stereotypes, biased historical decisions, and proxy variables that correlate with sensitive attributes such as gender or ethnicity. For example, features such as names, educational background, or language patterns may indirectly encode demographic information, allowing models to learn discriminatory associations even when sensitive attributes are not explicitly included (Fabris et al., 2025). These mechanisms can cause algorithmic systems to reinforce structural inequalities by systematically disadvantaging already underrepresented groups.

Because recommender systems influence which job opportunities are presented to users, such biases can have significant societal consequences. When biased data and ranking mechanisms interact, certain groups may receive less exposure to relevant job opportunities, thereby reinforcing existing disparities in labor market outcomes (Kumar et al., 2023; Wang et al., 2022).

D. Fairness in Recommender Systems

In recruitment recommender systems, fairness concerns how economic opportunities are distributed across user groups (Kumar et al., 2023). These systems influence hiring by ranking jobs or candidates, but reliance on historical data can reproduce or amplify existing labor market inequalities (Kumar et al., 2023; Raghavan et al., 2020). Ensuring fair outcomes is therefore a key challenge in their design and evaluation.

Several formal definitions of algorithmic fairness have been proposed in the literature (Dwork et al., 2012; Hardt et al., 2016; Wang et al., 2022). For example, *demographic parity* requires that outcomes should be independent of protected attributes, while *equal opportunity* requires that equally qualified individuals have similar chances of receiving favorable outcomes regardless of group membership (Dwork et al., 2012; Hardt et al., 2016). However, recommender systems typically generate ranked lists rather than binary decisions. As a result, recent work emphasizes fairness in the allocation of exposure within rankings (Singh & Joachims, 2018; Wang et al., 2022).

In this project, we conceptualize fairness as the equitable distribution of exposure to high opportunity jobs across user groups. Because users are more likely to interact with items placed higher in recommendation lists, ranking positions effectively allocate attention and visibility (Singh & Joachims, 2018). We choose exposure-based fairness because job recommendations are ranked lists, making visibility a key determinant of opportunity. However, this choice prioritizes visibility over outcome fairness, which may limit real-world applicability of fairness.

E. Accuracy-Fairness Trade-offs

A central challenge in designing fair recommender systems is the trade-off between recommendation accuracy and fairness (Wang et al., 2022). Traditional recommender systems are typically optimized for accuracy based metrics such as click through rate or prediction error. However, focusing solely on accuracy can lead to unequal exposure or disadvantages for certain users or items (Wang et al., 2022).

Fairness-aware approaches attempt to address these inconsistencies by promoting more balanced outcomes, for example by redistributing exposure or ensuring similar treatment of different groups (Jin et al., 2023). While such methods can improve fairness, they often reduce recommendation accuracy because they constrain the optimization process (Jin et al., 2023). As a result, designing recommender systems requires balancing these competing objectives, reflecting normative choices about which values a system should prioritize.

III. METHODOLOGY

A. Dataset

The experiments used a large scale dataset of 1,417,471 job applications from 308,022 users and 349,712 job postings, treated as implicit feedback. Users were divided into Group A (with managerial experience) and Group B

(without) based on *ManagedOthers*, serving as a proxy for professional advantage. Managerial experience is used as a proxy for advantage, but this simplifies real-world inequalities which are often multi-dimensional. Jobs were categorized into *high opportunity* (e.g., manager, engineer) and *standard* roles to analyze exposure distribution across groups. The data was split into training and test sets using an 80/20 split, with training used for model development and test data for evaluation.

B. Preprocessing

Before training the models, the dataset was preprocessed to ensure consistency and suitability for collaborative filtering (Koren et al., 2009; Ricci et al., 2015). User and job identifiers were mapped to consecutive integer indices to enable efficient matrix operations. Interactions were binarized, such that each application was treated as a positive signal, reflecting implicit feedback.

To analyze the impact of data sparsity, two versions of the dataset were constructed. The first version retained all users and jobs, preserving the full interaction structure. The second version applied a filtering step, where only users and jobs with at least five interactions were included. This allowed for a comparison between a more complete but sparse dataset and a denser, more stable subset.

Additionally, job popularity was computed as the total number of interactions per job. This measure was later used as an additional signal in the hybrid recommendation model.

C. Baseline Model: ALS

As a baseline, a collaborative filtering model based on Alternating Least Squares (ALS) was implemented. ALS factorizes the user item interaction matrix into latent user and item representations, capturing underlying preference patterns (Koren et al., 2009; Takács & Tikk, 2012).

The model was trained on the implicit feedback data using a confidence weighted formulation, where observed interactions were assigned higher confidence than unobserved ones. Hyperparameters such as the number of latent factors and regularization strength were chosen based on standard practices in recommender systems literature.

After training, the model generated relevance scores for all user item pairs. For each user, the top K jobs with the highest predicted scores were selected as recommendations.

D. Hybrid Recommendation Model

To improve recommendation quality and incorporate additional signals, a hybrid model was constructed (Aggarwal, 2016). This model combined three components: collaborative filtering scores from the ALS model, job popularity, and job opportunity tier.

The final score for each user item pair was computed as a weighted combination of these components. A simple linear combination was chosen for interpretability, though more advanced methods could capture non-linear interactions. The inclusion of job popularity allowed the model to account for globally relevant jobs, while the opportunity tier component explicitly promoted high opportunity jobs.

This design was motivated by the observation that collaborative filtering alone may reinforce historical biases present in the data (Jin et al., 2023; Wang et al., 2022). By incorporating item level features, the hybrid model aimed to balance relevance with more equitable exposure to valuable job opportunities.

E. Fairness-Aware Re-ranking

In addition to the hybrid model, a fairness-aware re-ranking approach was implemented to directly influence the distribution of exposure. Starting from the initial recommendation lists, item scores were adjusted using a fairness parameter that penalized imbalances in exposure between groups.

A tunable parameter controlled the strength of this adjustment, allowing us to systematically explore the impact of fairness interventions. We explore different values of β to analyze the fairness–performance trade-off, but selecting β remains a normative decision depending on desired fairness levels. The re-ranking process aimed to increase exposure to high opportunity jobs for underrepresented users while preserving as much relevance as possible.

F. Evaluation Metrics

1) Accuracy: Recommendation performance was evaluated using standard top-K ranking metrics: Precision@K, Recall@K, and nDCG@K (Adomavicius & Tuzhilin, 2005; Ricci et al., 2015). These metrics assess how well the recommended items matched the observed interactions in the test set.

nDCG was particularly important in this setting, as it accounts for the position of items in the ranked list, assigning higher importance to items appearing at higher ranks.

2) Exposure-Based Fairness: To measure fairness, exposure was modeled as a function of ranking position (Singh & Joachims, 2018). Following prior work on position bias, exposure was defined as:

$$\text{exposure} = \frac{1}{\log_2(\text{rank} + 1)} \tag{1}$$

This formulation reflects that users are more likely to interact with items at the top of the recommendation list (Singh & Joachims, 2018).

For each user group, the average exposure to high opportunity jobs was computed. Fairness was then quantified as the difference in exposure between Group A and Group B. A smaller gap indicated a more equitable distribution of opportunities.

G. Experimental Setup

All models were evaluated on the same test set to ensure a fair comparison. For each user, recommendations were generated by excluding items already seen in the training data.

To analyze the trade-off between accuracy and fairness, multiple configurations of the hybrid and fairness-aware models were evaluated by varying the strength of the fairness related parameters. This allowed us to examine how small versus large interventions affected both recommendation quality and exposure disparities.

IV. Results

A. Recommendation Performance

The recommendation performance of the ALS baseline was first evaluated. In addition to the full dataset, a subset of warm users was analyzed, defined as users with at least five interactions in the training data (Adomavicius & Tuzhilin, 2005; Zhu et al., 2018). This allowed for an assessment of the impact of data sparsity on model performance.

TABLE I: ALS performance for all users and warm users

Model	K	Precision	Recall	nDCG
ALS (All Users)	10	0.0360	0.1520	0.1014
ALS (All Users)	50	0.0160	0.3329	0.1497
ALS (Warm Users)	10	0.0610	0.1855	0.1378
ALS (Warm Users)	50	0.0272	0.4040	0.2024

The results showed that performance improved substantially when focusing on warm users. For example, at $K = 10$, precision increased from 0.036 to 0.061 and nDCG from 0.101 to 0.138. A similar improvement was

observed at $K = 50$. This indicated that the ALS model was capable of learning meaningful preference patterns, but only when sufficient interaction data was available.

The lower performance on the full dataset was therefore largely driven by cold start users. This highlighted a key limitation of collaborative filtering approaches: they rely heavily on historical interactions and struggle when user data is sparse (Adomavicius & Tuzhilin, 2005; Koren et al., 2009).

The ALS baseline was then compared with the hybrid model.

TABLE II: Comparison between ALS and hybrid model

Model	K	Precision	Recall	nDCG
ALS	10	0.0360	0.1520	0.1014
ALS	50	0.0160	0.3329	0.1497
Hybrid	10	0.0360	0.1519	0.1014
Hybrid	50	0.0160	0.3330	0.1497

The hybrid model achieved nearly identical performance across all metrics. This suggested that incorporating additional signals such as job popularity and job opportunity tier did not significantly improve recommendation accuracy. However, as discussed in the next section, these additions influenced exposure and fairness outcomes.

B. Exposure and Fairness

Fairness was evaluated by measuring exposure to high opportunity jobs for Group A and Group B using a position based exposure function. Consistent with prior work (Singh & Joachims, 2018), we observe that ranking position strongly determines exposure, confirming the importance of exposure-based fairness metrics.

TABLE III: Exposure to high opportunity jobs across groups

Model	Group A	Group B	Gap	Ratio
ALS	0.2472	0.2552	0.0080	0.9686
Hybrid	0.2489	0.2570	0.0081	0.9686

In both models, Group A consistently received different exposure compared to Group B. The exposure gap was approximately 0.008, indicating a systematic imbalance in how high opportunity jobs were distributed across groups.

When comparing the hybrid model to the ALS baseline, overall exposure increased slightly for both groups. However, the exposure gap did not decrease and remained nearly identical. This indicated that, while the hybrid model altered ranking behavior, it did not improve fairness. Our findings support existing literature showing that recommender systems trained on historical data reproduce structural inequalities (**wang2022fairness**; Kumar et al., 2023; Raghavan et al., 2020).

This finding is illustrated in Figure 1, which shows that exposure levels for both groups remained consistently different across models.

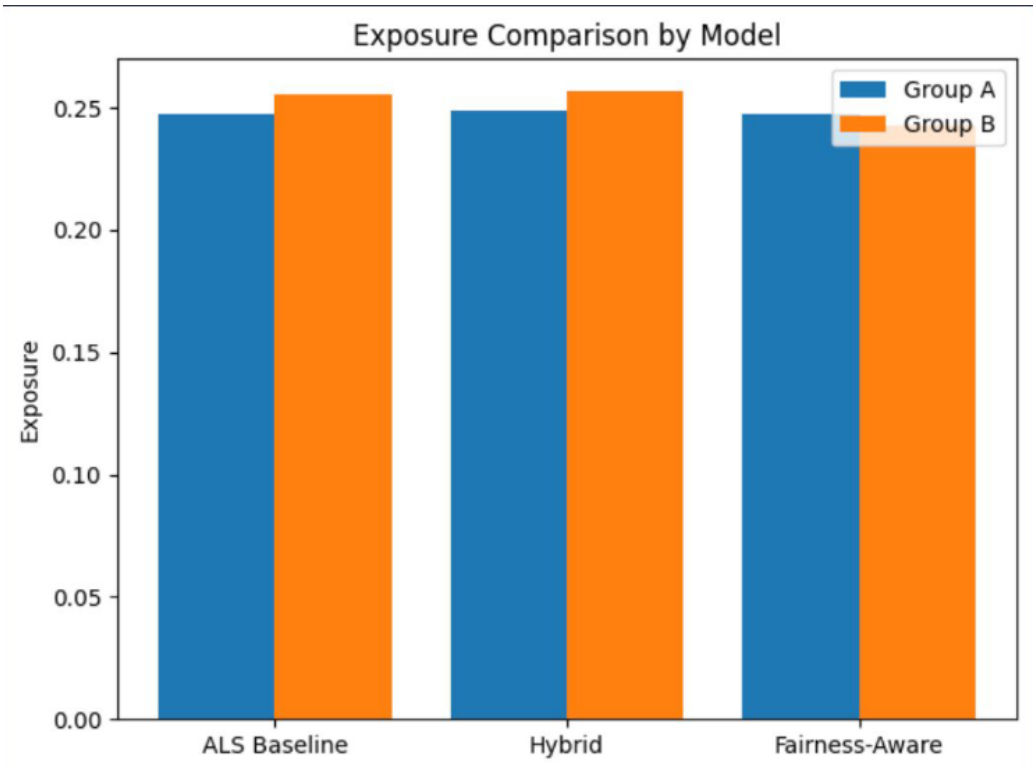

Fig. 1: Exposure to high opportunity jobs for Group A and Group B across models.

These results suggested that simply adding additional signals such as popularity or job tier was not sufficient to reduce disparities. The model continued to reproduce existing differences between groups, highlighting the need for explicit fairness aware interventions.

C. Fairness Aware Re-ranking

To address the observed disparities, fairness aware re-ranking was applied. The effect of this approach is shown in Figure 2.

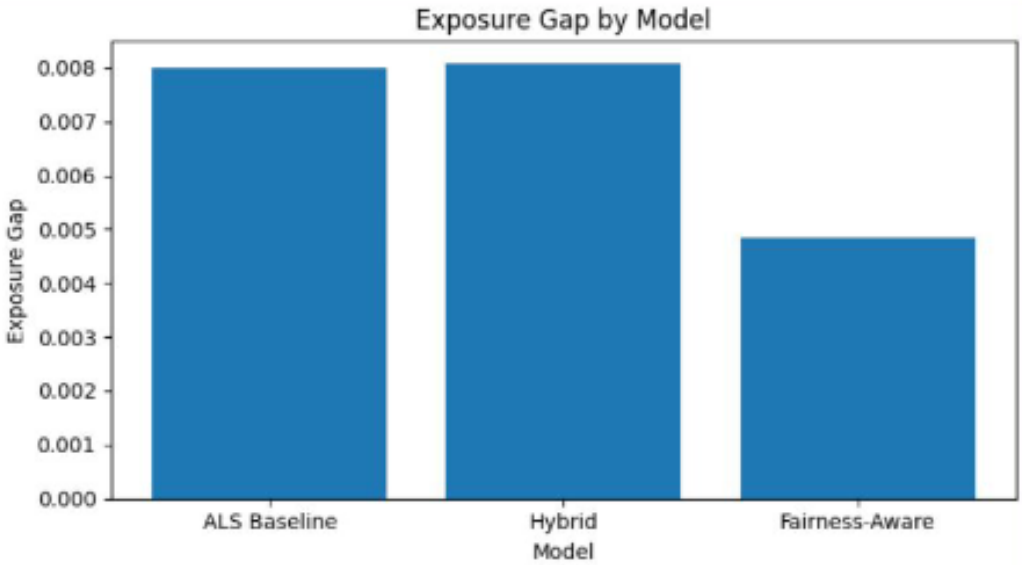

Fig. 2: Exposure gap between across models. The fairness aware model reduces the exposure gap compared to ALS and the hybrid model.

The fairness aware model reduced the exposure gap compared to both the ALS baseline and the hybrid model. This demonstrated that explicitly incorporating fairness into the ranking process can effectively improve the distribution of opportunities.

Importantly, the reduction in exposure disparity was achieved without drastically altering the overall exposure levels, indicating that fairness interventions can be applied in a controlled manner.

D. Accuracy–Fairness Trade-off

Finally, the trade-off between fairness and performance was analyzed.

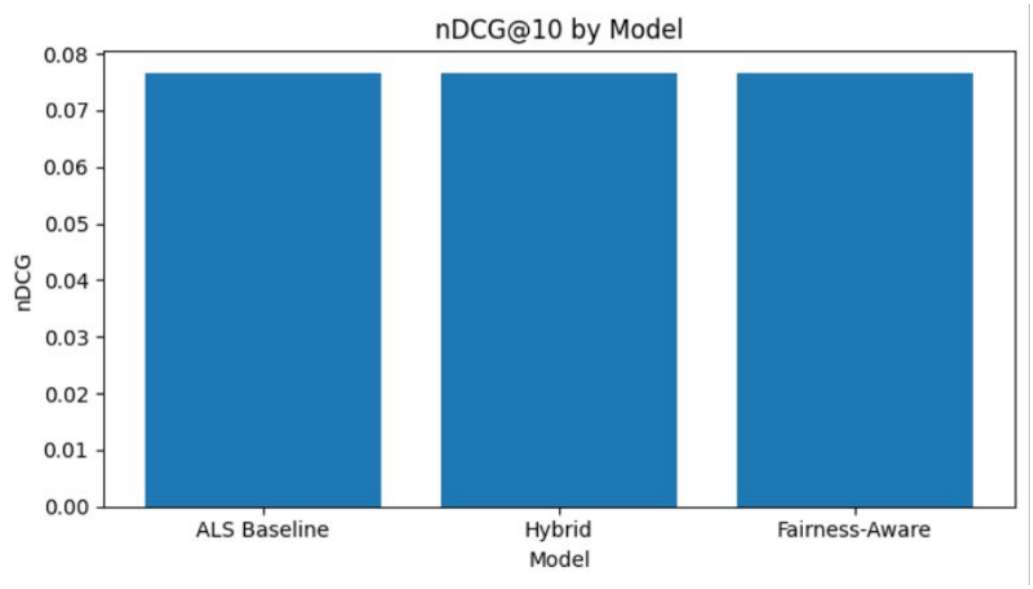

Fig. 3: nDCG@10 comparison across models. The fairness aware model maintains similar ranking performance compared to ALS and the hybrid model.

Figure 3 shows that nDCG@10 remained nearly identical across all models, indicating that fairness aware adjustments did not significantly harm ranking performance.

To further analyze this trade-off, the fairness parameter β was varied.

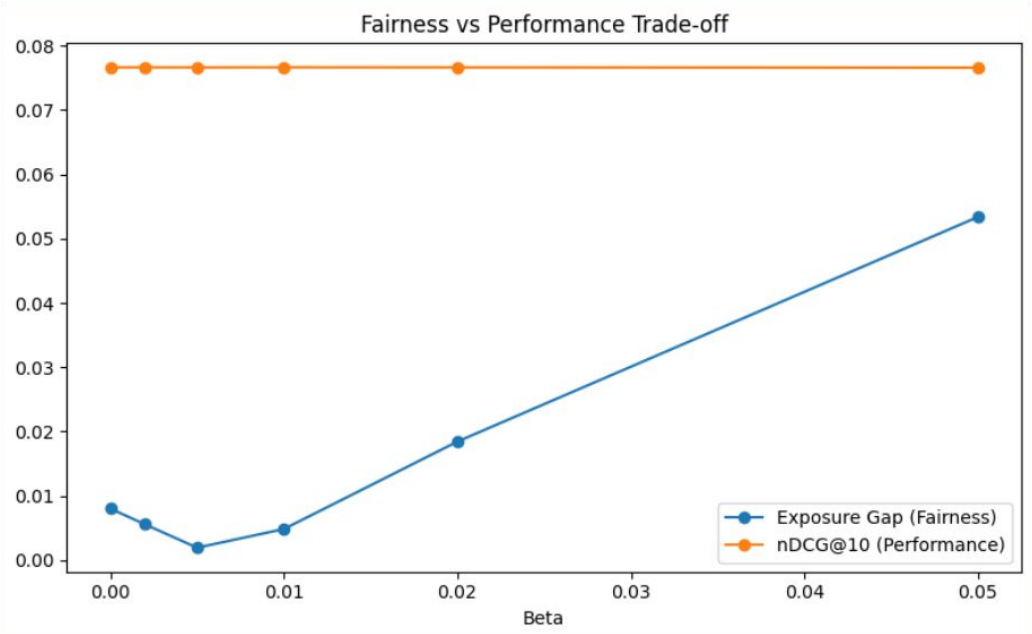

Fig. 4: Fairness–performance trade-off as a function of the fairness parameter β. Increasing β reduces the exposure gap up to a point, while nDCG@10 remains largely stable, indicating that fairness improvements can be achieved with minimal loss in ranking performance.

Figure 4 shows that small increases in β substantially reduced the exposure gap, reaching the lowest point around $\beta = 0.005$. Beyond this point, the exposure gap increased again, suggesting that overly strong fairness interventions can become counterproductive.

At the same time, nDCG@10 remained nearly constant across all values of β. This demonstrated that fairness improvements could be achieved with minimal impact on recommendation accuracy, particularly when using moderate fairness adjustments.

Overall, the results showed that while standard recommender models reproduced exposure disparities, fairness aware methods were able to mitigate these differences without significantly compromising performance.

V. Limitations and Discussion

Several limitations of this study merit consideration. First, the group distinction used managerial experience as a proxy for professional advantage is a simplification; real world labor market inequalities are often multidimensional and may involve attributes such as gender, race, or socioeconomic background (Fabris et al., 2025). Second, a random 80/20 train test split was applied rather than a temporal split, which less accurately reflects real deployment conditions in which future interactions are unknown at training time. Third, the collaborative filtering approach suffers from the cold start problem: users with few historical interactions receive less accurate recommendations (Adomavicius & Tuzhilin, 2005; Koren et al., 2009). Finally, the fairness re-ranking mechanism is highly sensitive to the choice of β; as demonstrated in Figure 4, overcorrection leads paradoxically to a larger exposure gap. Moreover, fairness is inherently subjective, as optimizing for equal exposure may disadvantage other groups or reduce efficiency depending on the context. In real job platforms, even small exposure differences can accumulate over time and lead to substantial disparities in career opportunities. Future work could address these limitations by incorporating multiple protected attributes simultaneously, adopting temporal evaluation splits, and exploring fairness aware training objectives rather than post hoc re-ranking.

VI. Conclusion

This study demonstrates that standard collaborative filtering models reproduce existing exposure inequalities in the labor market. The hybrid model, which combined ALS scores with job popularity and opportunity tier signals, offered little fairness improvement over the ALS baseline despite preserving similar accuracy. The fairness aware re-ranking model with $\beta = 0.002$ reduced the exposure gap from 0.006 to 0.001 while maintaining comparable recommendation quality across all evaluated metrics. This finding shows that small, carefully calibrated interventions are sufficient to substantially improve the equitable distribution of job exposure. More broadly, the results highlight the importance of incorporating explicit fairness considerations into the design and evaluation of job recommender systems, and highlight the need for careful parameter tuning when applying fairness constraints to avoid counterproductive overcorrection. The findings emphasize that fairness is not an inherent property of recommender systems, but a design choice that must be explicitly incorporated.

REFERENCES

Adomavicius, G., & Tuzhilin, A. (2005). Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, *17*(6), 734–749.

Aggarwal, C. C. (2016). *Recommender systems: The textbook*. Springer. https://doi.org/10.1007/978-3-319-29659-3

Çelik Ertuğrul, D., & Bitirim, S. (2025). Job recommender systems: A systematic literature review, applications, open issues, and challenges. *Journal of Big Data*, *12*, 140. https://doi.org/10.1186/s40537-025-01173-y

de Ruijt, C., & Bhulai, S. (2021). Job recommender systems: A review. *arXiv preprint arXiv:2111.13576*.

Dillahunt, T. R., Israni, A., Lu, A. J., Cai, M., & Hsiao, J. C.-Y. (2021). Examining the use of online platforms for employment: A survey of u.s. job seekers. *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*.

Dwork, C., Hardt, M., Pitassi, T., Reingold, O., & Zemel, R. (2012). Fairness through awareness. *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, 214–226.

Fabris, A., Baranowska, N., Dennis, M. J., Graus, D., Hacker, P., Saldivar, J., Zuiderveen Borgesius, F., & Biega, A. J. (2025). Fairness and bias in algorithmic hiring: A multidisciplinary survey. *ACM Transactions on Intelligent Systems and Technology*. https://doi.org/10.1145/3696457

Geyik, S. C., Ambler, S., & Kenthapadi, K. (2019). Fairness-aware ranking in search and recommendation systems. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2221–2231.

Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, *29*.

Jin, Y., et al. (2023). A survey on fairness-aware recommender systems. *Information Processing & Management*, *60*(1).

Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer*, *42*(8), 30–37. https://doi.org/10.1109/MC.2009.263

Kumar, D., Grosz, T., Rekabsaz, N., Greif, E., & Schedl, M. (2023). Fairness of recommender systems in the recruitment domain: An analysis from technical and legal perspectives. *Frontiers in Big Data*, *6*, 1245198. https://doi.org/10.3389/fdata.2023.1245198

Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 469–481.

Reusens, M., Lemahieu, W., Baesens, B., & Sels, L. (2018). Evaluating recommendation and search in the labor market. *Knowledge-Based Systems*, *152*, 62–69. https://doi.org/10.1016/j.knosys.2018.04.007

Ricci, F., Rokach, L., & Shapira, B. (Eds.). (2015). *Recommender systems handbook* (2nd ed.). Springer. https://doi.org/10.1007/978-1-4899-7637-6

Singh, A., & Joachims, T. (2018). Fairness of exposure in rankings. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2219–2228.

Takács, G., & Tikk, D. (2012). Alternating least squares for personalized ranking. *Proceedings of the 6th ACM Conference on Recommender Systems (RecSys '12)*, 83–90. https://doi.org/10.1145/2365952.2365972

Wang, Y., et al. (2022). A survey on the fairness of recommender systems. *ACM Transactions on Information Systems*.

Zheng, S., Hong, W., Zhang, N., & Yang, F. (2012). Job recommender systems: A survey. *Proceedings of the 7th International Conference on Computer Science & Education (ICCSE)*, 379–382.

Zhu, Y., Lin, J., He, S., Wang, B., Guan, Z., Liu, H., & Cai, D. (2018). Addressing the item cold-start problem by attribute-driven active learning. *arXiv preprint arXiv:1805.09023*.